BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

Module 4 Packet.

Respiratory · Lymphatic · Digestive · Lower Limb · Endocrine

PART A	**Course Notes** 5 chapters, in the order of your competency packet. Your reference for every video, lab, and exam question in Module 4.
PART B	**Lab Structure List** The Module 4 slice of the master lab list. Everything here is testable on the lab portion of the Module 4 exam.
ALSO	**Note Sheets** Your drawing pages for every Module 4 competency come as their own packet, BIO004–M4–Note-Sheet-Packet.

BIO 004 · MODULE 4

Contents

5 chapters, in the order of your competency packet. Part A of the Module 4 packet. Part B is the lab structure list.

1 **The Respiratory System**
RESPIRATORY · COMPETENCIES 97 TO 104

2 **The Lymphatic System**
LYMPHATIC · COMPETENCIES 105 TO 108

3 **The Alimentary Canal**
DIGESTIVE · COMPETENCIES 109 TO 119

4 **The Accessory Digestive Organs**
DIGESTIVE · COMPETENCIES 120 TO 125

5 **Endocrine Anatomy**
ENDOCRINE · COMPETENCIES 128 TO 132

Everything else for Module 4

Scan for the course home: the concept videos, Mastery OS, the calendar, and the exam schedule. Nothing behind this code is reprinted in this packet.

WHAT MODULE 4 COVERS

Module 4 is thirty six competencies, numbers 97 to 132 in your competency packet. Respiratory opens it, because it finishes the story the cardiovascular module started: blood leaves the heart, and this is where it is reloaded.

From there the module works through the systems that move material into the body and drain what is left behind. The lymphatic system collects what the capillaries leave in the tissues. The digestive system takes food apart, and its accessory organs supply the tools. Endocrine anatomy closes the module, because its glands sit inside the systems around it. In lab you also work the lower limb muscles and the vessels and nerves of the abdomen and lower limb (competencies 126 and 127), which are lab only and live in Part B.

This is anatomy. Where function appears in these notes it is there because the structure does not make sense without it.

PART A

Course Notes.

5 chapters, in the order of your competency packet. Read a chapter before the class day it belongs to, then draw its competencies on your note sheets.

1 **The Respiratory System** (competencies 97 to 104)

2 **The Lymphatic System** (competencies 105 to 108)

3 **The Alimentary Canal** (competencies 109 to 119)

4 **The Accessory Digestive Organs** (competencies 120 to 125)

5 **Endocrine Anatomy** (competencies 128 to 132)

CHAPTER 1

The Respiratory System

THE RESPIRATORY SYSTEM

COMPETENCIES 97 TO 104

A reference for the respiratory system video and lab. This chapter covers the upper and lower respiratory tracts, the larynx, the bronchial tree, the lungs and pleurae, the alveoli, and common respiratory disorders. The focus is on structure.

BY THE END

1. Distinguish the upper and lower respiratory tracts and the conducting and respiratory zones.
2. Identify the parts of the nasal cavity and the pharynx.
3. Identify the cartilages of the larynx and the parts of the bronchial tree.
4. Describe the lungs, the pleurae, and the structure of an alveolus.
5. Name common respiratory disorders.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The respiratory system

drsrennie-stack.github.io/new-build-bio4-solano/respiratory-concept-videos.html

The Respiratory System, an Overview

The respiratory system moves air in and out and brings it close to the blood. Its parts are grouped two ways, and the two groupings do not line up, which is exactly why both are tested.

THE TWO WAYS OF GROUPING THE PARTS

Structure	What it is
Respiratory system	The organs that move air and bring it close to the blood for gas exchange
Upper respiratory tract	The nose, nasal cavity, paranasal sinuses, and pharynx
Lower respiratory tract	The larynx, trachea, bronchi, and lungs
Conducting zone	The passages that only carry air, from the nose to the terminal bronchioles. It warms, moistens, and filters
Respiratory zone	The structures where gas exchange happens: the respiratory bronchioles, alveolar ducts, and alveoli
A note on scope	This chapter covers the structure of the system. The exchange of gases is a physiology topic and is not covered here

The Upper Respiratory Tract

The upper tract takes in air and conditions it, warming, moistening, and filtering it before it reaches the lungs.

THE NOSE AND NASAL CAVITY

Structure	What it is

Structure	What it is
External nose	The visible part, framed by the nasal bones and the frontal process of the maxilla above and hyaline cartilage below
Root	The area between the eyebrows, where the nose meets the forehead
Bridge	The upper bony part of the nose
Dorsum nasi	The anterior margin, running from root to apex
Apex	The tip of the nose
Nares	The external openings, the nostrils
Nasal cavity	The space behind the nose, divided into right and left halves by the nasal septum
Nasal septum	Septal cartilage in front, the vomer and the perpendicular plate of the ethmoid behind
Nasal vestibule	The area just inside the nostrils, lined with skin, sebaceous glands, and coarse hairs that filter particles
Nasal conchae	The three scroll-shaped projections on the lateral wall that swirl the air and increase contact with the warm, moist lining
Nasal meatuses	The passages beneath each concha, named superior, middle, and inferior
Hard palate	The bony floor of the nasal cavity, separating it from the oral cavity
Soft palate	The muscular posterior floor, which lifts to close the nasopharynx during swallowing
Choanae	The posterior openings where the nasal cavity becomes the nasopharynx
Paranasal sinuses	Air-filled cavities in the frontal, ethmoid, sphenoid, and maxillary bones that drain into the nasal cavity

The mucosa of the nasal cavity

TWO MUCOSAE, TWO JOBS

Mucosa	Epithelium	Role
Olfactory mucosa	Olfactory epithelium, in a small patch in the roof	Houses the receptors for smell
Respiratory mucosa	Pseudostratified ciliated columnar with goblet cells	Secretes mucus, traps particles, and sweeps them toward the pharynx to be swallowed

HOLD ONTO THIS

Cold air makes the cilia sluggish, so the mucus stops moving backward and runs forward instead. That is why a nose runs in cold weather, and it is a structural answer to an everyday question.

The pharynx

THE THREE REGIONS, TOP TO BOTTOM

Region	Location	Carries	Epithelium and structures
Nasopharynx	Behind the nasal cavity, above the soft palate	Air only	Pseudostratified ciliated columnar. Holds the pharyngeal tonsil and the openings of the auditory tubes
Oropharynx	Behind the oral cavity, from the soft palate to the epiglottis	Air and food	Stratified squamous, for friction. Holds the palatine and lingual tonsils
Laryngopharynx	From the epiglottis to the esophagus	Air and food	Stratified squamous. The point where the air and food paths divide

The Larynx

The larynx, the voice box, connects the pharynx to the trachea. It keeps food out of the airway, holds the airway open, and houses the vocal cords. Nine cartilages build it, three single and three paired.

THE NINE CARTILAGES

Cartilage	Number	Description
Thyroid cartilage	Single	The large shield-shaped cartilage forming the laryngeal prominence, the Adam's apple
Cricoid cartilage	Single	The ring-shaped cartilage just below the thyroid cartilage, the only complete ring in the airway
Epiglottis	Single	The flexible spoon-shaped flap of elastic cartilage that folds over the laryngeal inlet during swallowing
Arytenoid cartilages	Paired	Anchor the vocal folds and move them. The most important of the paired cartilages
Corniculate cartilages	Paired	Sit on top of the arytenoids
Cuneiform cartilages	Paired	Lie in the fold between the arytenoid and the epiglottis

THE FOLDS AND THE OPENING

Structure	What it is
Vocal folds	The true vocal cords. Pearly white and avascular, they vibrate as air passes to produce sound
Vocal ligaments	The elastic bands within the vocal folds, running from the arytenoid cartilages to the thyroid cartilage
Vestibular folds	The false vocal cords, above the true folds. They close the airway during swallowing and play no part in sound
Glottis	The vocal folds together with the opening between them

HOLD ONTO THIS

Every laryngeal cartilage is hyaline except the epiglottis, which is elastic. It has to bend and spring back with every swallow, and hyaline cartilage would not.

The Trachea and Bronchial Tree

Below the larynx the airway becomes a branching tree, growing narrower and more numerous at every division. Follow the path of air in order.

1. **Trachea** the windpipe, held permanently open by C-shaped rings of hyaline cartilage, with the trachealis muscle closing the gap posteriorly against the esophagus
2. **Carina** the internal ridge where the trachea forks into the two main bronchi
3. **Main bronchi** the primary bronchi, one entering each lung. The right is wider and more vertical
4. **Lobar bronchi** the secondary bronchi, one to each lobe of the lung
5. **Segmental bronchi** the tertiary bronchi, one to each bronchopulmonary segment
6. **Bronchioles** the smallest conducting airways, with smooth muscle walls and no cartilage
7. **Terminal bronchioles** the last airways of the conducting zone, just before gas exchange begins

HOLD ONTO THIS

Cartilage decreases and smooth muscle increases as the tree gets smaller. The switch point is the bronchiole, which is why bronchioles can constrict and bronchi cannot, and why asthma is a disease of the small airways.

The Lungs and Pleurae

The two lungs fill most of the thoracic cavity. They are not identical: the left lung gives up room for the heart.

COMPARE THE TWO LUNGS

Lung	Lobes	Fissures	Note
Right lung	Three: superior, middle, and inferior	A horizontal fissure and an oblique fissure	The larger of the two
Left lung	Two: superior and inferior	An oblique fissure only	Smaller, with a cardiac notch that makes room for the heart

THE SURFACES AND THE COVERING

Structure	What it is
Apex	The narrow superior tip of a lung, rising just above the clavicle
Base	The broad inferior surface of a lung, resting on the diaphragm
Hilum	The indentation on the medial surface where the bronchus and vessels enter and leave
Root of the lung	The bundle at the hilum: main bronchus, pulmonary artery, pulmonary veins, bronchial vessels, nerves, and lymphatics
Parietal pleura	The serous membrane lining the thoracic wall
Visceral pleura	The serous membrane covering the outer surface of the lung
Pleural cavity	The thin, fluid-filled space between the two pleurae that lets the lung glide as it expands
Cardiac notch	The indentation in the left lung that accommodates the heart
Lingula	The tongue-shaped projection of the left superior lobe, below the cardiac notch

The Respiratory Zone and the Alveoli

Past the terminal bronchioles the airway becomes the respiratory zone, where the walls thin out into air sacs and gas exchange becomes possible.

THE STRUCTURES OF THE RESPIRATORY ZONE

Structure	What it is
Respiratory bronchioles	The first airways of the respiratory zone, with a few alveoli budding from their walls
Alveolar ducts	Passages whose walls are lined entirely with alveoli
Alveolar sac	A cluster of alveoli that share a common space
Alveoli	The tiny air sacs that are the site of gas exchange. The lungs hold hundreds of millions of them
Type I alveolar cells	The thin, flat simple squamous cells that form most of the alveolar wall, where gases cross
Type II alveolar cells	The cuboidal cells that secrete surfactant, which lowers surface tension and keeps the alveoli from collapsing
Alveolar macrophages	The dust cells that patrol the alveolar surface and engulf debris
Respiratory membrane	The thin barrier between the alveolar air and the blood: alveolar epithelium, fused basement membranes, and capillary endothelium
Pulmonary capillaries	The dense net of capillaries wrapping each alveolus

Respiratory Disorders

Respiratory disorders mostly block airflow or damage the surface where gases cross.

COMPARE THE COMMON ONES BY THE STRUCTURE EACH ONE AFFECTS

Disorder	Structure affected	What it is
Asthma	The bronchioles	Inflammation and narrowing of the bronchioles, causing wheezing and breathlessness
COPD	The airways and alveoli	Chronic obstructive pulmonary disease, a long-term blockage of airflow that includes emphysema and chronic bronchitis
Emphysema	The alveoli	Destruction of the alveolar walls, which reduces the surface available for gas exchange
Pneumonia	The alveoli	An infection that inflames the alveoli and fills them with fluid
Pneumothorax	The pleural cavity	Air in the pleural cavity, which breaks the seal and lets the lung collapse
Lung cancer	The lung tissue	A malignant tumor of the lung, strongly linked to smoking

WHY THE DESIGN HAS WEAK POINTS

The **epiglottis** normally seals the airway during a swallow. When something slips past it, a coughing fit is the airway's attempt to clear it. Because the right main bronchus is wider and more vertical than the left, an inhaled object most often lands in the right lung.

The **alveoli** give the lungs an enormous surface, but it is fragile. Emphysema destroys alveolar walls and merges many small sacs into a few large ones, trading a huge exchange surface for a small one. The change is structural and permanent.

The **pleural cavity** is sealed and holds only a thin film of fluid. If the chest wall or lung is punctured, air enters that space, the seal is lost, and the lung collapses, a pneumothorax. The structure that lets the lung glide is also what makes it vulnerable.

The **cricoid cartilage** is the only complete ring in the airway, which is why the space just above it, the cricothyroid membrane, is the landmark for an emergency airway. Anatomy chooses that spot, not convenience.

CHAPTER 2

The Lymphatic System

THE LYMPHATIC SYSTEM

COMPETENCIES 105 TO 108

A reference for the lymphatic system video and lab. This chapter covers lymph and the pathway it follows, the lymphatic vessels, lymph nodes, the lymphatic organs, and common disorders of the system.

BY THE END

1. Define lymph and trace the pathway it follows back to the blood.
2. Describe the structure of lymphatic vessels and a lymph node.
3. Sort the lymphatic organs into primary and secondary.
4. Name common disorders of the lymphatic system.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The lymphatic system

drsrennie-stack.github.io/new-build-bio4-solano/lymphatic-concept-videos.html

The Lymphatic System, an Overview

The lymphatic system is a one-way drainage network. It collects fluid that leaks out of the blood capillaries, filters it, and returns it to the bloodstream, while also housing the body's defenses.

WHAT THE SYSTEM IS AND WHAT IT DOES

Structure	What it is
Lymphatic system	A network of vessels, tissues, and organs that drains tissue fluid and defends the body
Interstitial fluid	The fluid in the spaces between cells, filtered out of the blood capillaries
Lymph	The interstitial fluid once it has entered the lymphatic vessels; a clear, watery fluid
Three functions	Draining excess interstitial fluid back to the blood, absorbing dietary fats from the small intestine, and housing the immune defenses
A one-way system	Lymphatic vessels carry lymph in one direction only, toward the heart; the system has no pump of its own

The Pathway of Lymph

Lymph travels a fixed route from the tissue spaces back to the bloodstream, growing into larger and larger vessels along the way. Follow it in order.

1. **Lymphatic capillaries** blind-ended tubes that take up excess interstitial fluid from the tissue spaces
2. **Lymphatic collecting vessels** larger vessels, with valves, that carry the lymph onward
3. **Lymph nodes** the lymph is filtered as it passes through nodes spaced along the vessels
4. **Lymphatic trunks** collecting vessels merge into large trunks, each draining a major region of the body
5. **Lymphatic ducts** the trunks empty into two ducts: the right lymphatic duct and the larger thoracic duct
6. **Subclavian veins** the ducts return the lymph to the bloodstream where the subclavian and internal jugular veins meet

Lymphatic Vessels

Lymphatic vessels resemble veins, but they begin as blind-ended capillaries and they move fluid without a pump.

THE VESSELS, FROM THE SMALLEST TO THE DUCTS

Structure	What it is
Lymphatic capillaries	The smallest lymphatic vessels, blind-ended and more permeable than blood capillaries
Overlapping endothelial cells	The cells of a lymphatic capillary overlap as one-way flap valves, letting fluid in but not back out
Lacteals	Specialized lymphatic capillaries in the lining of the small intestine that absorb dietary fat
Lymphatic valves	One-way valves in the collecting vessels that keep lymph moving in a single direction
How lymph moves	With no pump, lymph is pushed along by the squeeze of nearby skeletal muscles and by breathing movements
Right lymphatic duct	The smaller duct, draining lymph from the upper right quadrant of the body
Thoracic duct	The larger duct, draining the rest of the body; it begins at a sac called the cisterna chyli

Lymph Nodes

Lymph nodes are the filters of the system. Hundreds of them lie along the lymphatic vessels, cleaning the lymph before it returns to the blood.

THE PARTS OF A NODE AND THE VESSELS THAT SERVE IT

Structure	What it is
Lymph node	A small, bean-shaped organ that filters lymph and houses immune cells
Capsule	The dense connective tissue covering of a node
Cortex	The outer region of a node, packed with lymphocytes in rounded clusters
Medulla	The inner region of a node
Afferent vessels	The lymphatic vessels that carry lymph into a node
Efferent vessels	The lymphatic vessels that carry filtered lymph out of a node, leaving at the hilum
Node clusters	Nodes are especially concentrated in the cervical, axillary, and inguinal regions

HOLD ONTO THIS

Afferent vessels arrive, efferent vessels exit. The same pair of words names flow in and flow out everywhere in the body, so learning it once here pays off again in the kidney and the nervous system.

The Lymphatic Organs

The lymphatic organs are sorted into primary organs, where lymphocytes form and mature, and secondary organs, where lymphocytes meet and respond to invaders. Compare them.

THE PRIMARY AND SECONDARY LYMPHATIC ORGANS

Organ	Group	Location	Role
Red bone marrow	Primary	The spongy bone of the axial skeleton and limb girdles	Where all lymphocytes form, and where B cells mature
Thymus	Primary	The mediastinum, behind the sternum	Where T cells mature
Lymph nodes	Secondary	Clustered along the lymphatic vessels	Filter the lymph
Spleen	Secondary	The upper left abdomen, behind the stomach	Filters the blood
Tonsils	Secondary	A ring around the entrance to the throat	Guard against inhaled and swallowed pathogens
MALT	Secondary	The mucous membranes of the digestive, respiratory, and other tracts	Guards the body's mucosal surfaces

HOLD ONTO THIS

Only two organs are primary, the red bone marrow and the thymus, because those are the only places lymphocytes are made and matured. Everything else on the list is secondary, a place where a finished lymphocyte meets what it was built to fight.

The Spleen, Thymus, and Tonsils

Three of the lymphatic organs are worth a closer look at their structure.

The spleen

THE SPLEEN AND ITS TWO TISSUES

Structure	What it is
Spleen	The largest lymphatic organ, in the upper left abdomen behind the stomach
White pulp	The lymphocyte-rich tissue of the spleen that carries out its immune functions
Red pulp	The tissue of the spleen that filters the blood and removes worn-out erythrocytes

The thymus and tonsils

THE THYMUS AND THE RING OF TONSILS

Structure	What it is
Thymus	The primary organ where T cells mature; large in childhood and shrinking after puberty
Tonsils	A ring of lymphatic tissue guarding the throat, including the palatine, lingual, and pharyngeal tonsils
Pharyngeal tonsil	The single tonsil on the posterior wall of the nasopharynx, also called the adenoid

Disorders of the Lymphatic System

When drainage fails or the lymphatic tissue is diseased, the problems show up as swelling or as changes in the nodes and organs. Compare the common ones.

COMPARE THE COMMON ONES BY THE STRUCTURE EACH ONE AFFECTS

Disorder	Structure affected	What it is
Edema	The interstitial space	Swelling from excess interstitial fluid when lymphatic drainage cannot keep up
Lymphedema	The lymphatic vessels	Long-term tissue swelling from blocked or missing lymphatic vessels, sometimes after lymph node removal
Lymphadenopathy	The lymph nodes	Enlarged lymph nodes, often a sign the nodes are fighting an infection
Lymphoma	Lymphatic tissue	A cancer arising in lymphatic tissue, including Hodgkin and non-Hodgkin lymphoma
Splenomegaly	The spleen	An enlarged spleen, which can follow an infection or a blood disorder
Tonsillitis	The tonsils	Inflammation of the tonsils, common in childhood

THE SYSTEM A CLINICIAN READS THROUGH THE SKIN

Swollen glands are **lymphadenopathy**: lymph nodes enlarge as they trap and fight pathogens, which is why a clinician feels for tender nodes in the neck, armpit, and groin during an exam.

Because lymph passes through nodes in order, the lymphatic system also maps how some cancers spread. The first node a region drains into is the **sentinel node**, and checking it tells a surgeon whether a cancer has begun to travel. Removing many nodes can block drainage and leave lasting lymphedema in the limb.

The **spleen** sits just under the lower left ribs. Its soft tissue is easily torn, so a blow to that area can rupture the spleen and cause heavy internal bleeding, since the organ filters a large volume of blood.

CHAPTER 3

The Alimentary Canal

THE DIGESTIVE SYSTEM, THE ALIMENTARY CANAL

COMPETENCIES 109 TO 119

Covers the wall of the digestive tube, the peritoneum, and the regions of the canal from the mouth to the anus, with the named features of the stomach and the intestines. The focus is on the structures and the job each one does.

BY THE END

1. Distinguish the alimentary canal from the accessory organs and name the four layers of its wall.
2. Trace the pathway of food from the mouth to the anus.
3. Identify the named features of the stomach, small intestine, and large intestine.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The digestive system

drsrennie-stack.github.io/new-build-bio4-solano/gi-concept-videos.html

The Alimentary Canal, an Overview

The digestive system has two groups of organs. The alimentary canal is the continuous tube that food travels through; the accessory organs sit alongside it and empty their secretions in. Covers the canal.

THE TWO GROUPS OF DIGESTIVE ORGANS

Structure	What it is
Digestive system	The organs that take in food and eliminate waste; the alimentary canal plus the accessory organs
Alimentary canal	The gastrointestinal tract; a continuous muscular tube from the mouth to the anus
Organs of the canal	The mouth, most of the pharynx, the esophagus, the stomach, the small intestine, and the large intestine
Accessory organs	The teeth, tongue, salivary glands, liver, gallbladder, and pancreas; covered on their own sheet
Lumen	The open interior space of the tube, continuous with the outside world at the mouth and anus
Length	About 16 to 23 feet in a living person, longer once muscle tone is lost

The Wall of the Alimentary Canal

From the lower esophagus to the anal canal, the wall of the tube is built from the same four layers, deep to superficial. Compare them.

THE FOUR LAYERS OF THE WALL

Layer	What it contains
Mucosa	The inner lining; an epithelium in contact with food, the lamina propria of areolar connective tissue beneath it, and a thin muscularis mucosae
Submucosa	Areolar connective tissue with blood and lymphatic vessels, and the submucosal nerve plexus
Muscularis	Usually an inner circular and an outer longitudinal layer of smooth muscle, with the myenteric nerve plexus between them
Serosa	The outermost layer; a serous membrane, also called the visceral peritoneum, replaced by a fibrous adventitia on the esophagus

Two nerve networks in the wall make up the enteric nervous system, the nerve supply of the gut.

THE ENTERIC NERVOUS SYSTEM

Structure	What it is
Submucosal plexus	The nerve network in the submucosa that controls the secretions of the gut
Myenteric plexus	The nerve network between the muscle layers that controls the movement of the gut

HOLD ONTO THIS

Four layers everywhere, with two exceptions worth holding. The esophagus wears a fibrous adventitia where the rest of the tube has a serosa, and the stomach carries a third layer of muscle where the rest carries two.

The Peritoneum

The peritoneum is the large serous membrane of the abdomen. Its folds anchor the digestive organs and carry their vessels and nerves.

THE LAYERS OF THE PERITONEUM

Structure	What it is
Parietal peritoneum	The layer that lines the wall of the abdominal cavity
Visceral peritoneum	The layer that covers the organs; it is the serosa of those organs
Peritoneal cavity	The slim, fluid-filled space between the parietal and visceral layers
Retroperitoneal organs	Organs behind the peritoneum, not within its cavity; the kidneys, pancreas, duodenum, and parts of the colon

Five large folds of peritoneum weave among the organs. Compare what each one connects.

THE FIVE FOLDS OF PERITONEUM

Fold	What it connects
Greater omentum	The largest fold; a fatty apron hanging from the stomach over the intestines
Lesser omentum	Connects the stomach and duodenum to the liver, and carries vessels to the liver
Falciform ligament	Attaches the liver to the anterior abdominal wall and the diaphragm
Mesentery	A fan-shaped fold binding the jejunum and ileum to the posterior abdominal wall
Mesocolon	Binds the transverse and sigmoid colon to the posterior abdominal wall

The Mouth, Pharynx, and Esophagus

Food enters at the mouth, is swallowed through the pharynx, and travels the esophagus to the stomach.

FROM THE ORAL CAVITY TO THE STOMACH

Structure	What it is
Oral cavity	The mouth; bounded by the cheeks, the lips, the palate, and the tongue
Hard palate	The bony anterior roof of the mouth, formed by the maxillae and palatine bones
Soft palate	The muscular posterior roof of the mouth
Uvula	The finger-like projection of the soft palate that closes off the nasopharynx during swallowing
Fauces	The opening from the mouth into the oropharynx
Palatine and lingual tonsils	Lymphatic tissue between the palatal arches and at the base of the tongue
Pharynx	The funnel-shaped passage with three regions: the nasopharynx, oropharynx, and laryngopharynx
Esophagus	The collapsible muscular tube, about 10 inches long, that carries food from the laryngopharynx to the stomach
Esophageal hiatus	The opening in the diaphragm that the esophagus passes through
Upper and lower esophageal sphincters	The rings of muscle that guard the top and bottom of the esophagus; the lower is also called the cardiac sphincter

The Stomach

The stomach is a J-shaped widening of the tube below the diaphragm. It is a mixing chamber and a holding reservoir.

THE REGIONS AND NAMED FEATURES OF THE STOMACH

Structure	What it is
Cardia	The region that surrounds the opening from the esophagus
Fundus	The rounded portion that bulges above and to the left of the cardia
Body	The large central portion of the stomach
Pyloric part	The region that connects the body to the duodenum; it has the pyloric antrum, pyloric canal, and pylorus
Pyloric sphincter	The ring of smooth muscle between the pylorus and the duodenum
Lesser curvature	The concave superior border of the stomach
Greater curvature	The convex inferior border of the stomach
Rugae	The folds of the inner stomach lining that flatten as the stomach fills
Gastric pits and glands	The pits in the stomach lining that lead to the gastric glands, the secreting glands of the stomach
Muscularis of the stomach	Three layers of smooth muscle, an inner oblique, a middle circular, and an outer longitudinal; one more than the rest of the tube

The Small Intestine

The small intestine is the long, coiled region where most absorption happens. Its lining is folded again and again to expand the surface area.

THE REGIONS OF THE SMALL INTESTINE AND THE FOLDS OF ITS LINING

Structure	What it is
Duodenum	The first and shortest region, about 12 inches, curving from the pyloric sphincter; mostly retroperitoneal
Jejunum	The middle region, about 3 feet long
Ileum	The longest region, about 6 feet, joining the large intestine at the ileocecal sphincter
Ileocecal sphincter	The valve of smooth muscle where the ileum meets the large intestine
Circular folds	The plicae circulares; permanent ridges of the mucosa and submucosa that increase surface area
Villi	Finger-like projections of the mucosa that give the lining a velvety look and greatly increase surface area
Lacteal	The lymphatic capillary inside each villus that absorbs dietary fats
Microvilli	The microscopic projections of the absorptive cells that form the brush border

HOLD ONTO THIS

The small intestine amplifies its surface three times over, each fold smaller than the last: circular folds of the wall, villi of the mucosa, and microvilli on the single absorptive cell.

The Large Intestine

The large intestine is the wider, final region of the canal. It absorbs water and forms and stores the feces.

THE REGIONS AND NAMED FEATURES OF THE LARGE INTESTINE

Structure	What it is
Cecum	The pouch at the start of the large intestine, just below the ileocecal valve
Vermiform appendix	The narrow, worm-like tube of lymphatic tissue attached to the cecum
Colon	The long portion with four parts: the ascending, transverse, descending, and sigmoid colon
Rectum	The straight portion, about 6 inches, anterior to the sacrum and coccyx
Anal canal	The last 2 to 3 cm of the tract, ending at the anus
Internal and external anal sphincters	The involuntary smooth muscle and voluntary skeletal muscle that guard the anus
Teniae coli	Three thickened bands of longitudinal muscle running the length of the colon
Haustra	The pouches that give the colon its puckered appearance, formed by the teniae coli
Omental appendices	Small fat-filled pouches of peritoneum attached to the teniae coli

The Pathway of Food

Food moves through the alimentary canal in one direction, through each region in turn.

1. **Mouth** food is taken in, chewed, and mixed with saliva
2. **Pharynx** the swallowed food passes through the oropharynx and laryngopharynx
3. **Esophagus** muscular contractions carry the food down to the stomach
4. **Stomach** food is churned and held as it becomes a liquid
5. **Small intestine** the duodenum, jejunum, and ileum, where most absorption occurs
6. **Large intestine** the cecum, colon, rectum, and anal canal, where water is absorbed and feces form
7. **Anus** the feces are eliminated

Common Disorders of the Alimentary Canal

Compare the common disorders of the canal by the structure each one affects.

THE COMMON DISORDERS OF THE CANAL

Disorder	What it is
Gastroesophageal reflux disease (GERD)	Stomach contents flow back up through a weak lower esophageal sphincter and irritate the esophagus
Hiatal hernia	Part of the stomach pushes up through the esophageal hiatus of the diaphragm
Peptic ulcer	An eroded sore in the wall of the stomach or duodenum
Appendicitis	Inflammation of the vermiform appendix; rupture can spread infection to the peritoneum
Diverticulosis	Small pouches that bulge outward through weak spots in the colon wall
Inflammatory bowel disease	Chronic inflammation of the intestinal wall, as in Crohn disease and ulcerative colitis
Hemorrhoids	Swollen, dilated veins in the wall of the anal canal and rectum
Colorectal cancer	A malignant tumor of the colon or rectum

THE LOCATION OF A PROBLEM NAMES THE SYMPTOM

Because each region of the canal has its own job, where a problem sits predicts what the patient feels. A weak **lower esophageal sphincter** brings heartburn behind the breastbone; an inflamed **appendix** brings pain to the lower right abdomen.

The four-layer wall explains how a problem deepens. A **peptic ulcer** begins in the mucosa, and the danger is that it erodes through the submucosa and muscularis. If it reaches the serosa and perforates, the contents of the lumen spill into the peritoneal cavity.

CHAPTER 4

The Accessory Digestive Organs

ACCESSORY ORGANS OF DIGESTION

COMPETENCIES 120 TO 125

Covers the teeth and tongue, the salivary glands, the liver, the gallbladder and the path of bile, and the pancreas. The focus is on the structures and the job each one does.

BY THE END

1. Name the six accessory digestive organs and the role of each.
2. Identify the structures of the liver, gallbladder, and pancreas.
3. Trace the path of bile from the liver to the small intestine.



WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Accessory digestive organs

drsrennie-stack.github.io/new-build-bio4-solano/accessory-digestive-concept-videos.html

The Accessory Organs, an Overview

The accessory digestive organs are not part of the food tube itself. The teeth and tongue work on food in the mouth; the rest add secretions to the canal through ducts.

THE SIX ACCESSORY ORGANS AND HOW THEY WORK

Structure	What it is
Accessory digestive organs	The teeth, tongue, salivary glands, liver, gallbladder, and pancreas
Teeth and tongue	The accessory organs that make direct contact with food in the mouth
Glandular accessory organs	The salivary glands, liver, and pancreas, which produce secretions delivered through ducts
Storage organ	The gallbladder, which stores and concentrates bile until it is needed

The Teeth and Tongue

The teeth and the tongue prepare food in the mouth, breaking it down and moving it for chewing and swallowing.

THE STRUCTURES OF THE TEETH AND THE TONGUE

Structure	What it is
Teeth	The accessory organs that cut and grind food during chewing
Tongue	A digestive organ of skeletal muscle covered with mucous membrane; it forms the floor of the mouth
Extrinsic tongue muscles	Muscles that originate outside the tongue and move it side to side and in and out
Intrinsic tongue muscles	Muscles within the tongue that alter its shape and size for speech and swallowing
Lingual frenulum	The midline fold of mucous membrane anchoring the tongue to the floor of the mouth
Papillae	The projections on the tongue surface that carry taste buds and aid the grip on food
Lingual glands	Small glands in the tongue that secrete mucus and watery fluid

The Salivary Glands

Saliva keeps the mouth moist, cleanses it, and begins the breakdown of food. Three large pairs of glands produce most of it. Compare them.

THE THREE LARGE PAIRS OF SALIVARY GLANDS

Gland	Location
Parotid glands	Anterior and inferior to the ears, between the skin and the masseter muscle; the duct opens by the second upper molar
Submandibular glands	In the floor of the mouth, along the inner side of the mandible; the ducts open near the lingual frenulum
Sublingual glands	Beneath the tongue, above the submandibular glands; the ducts open onto the floor of the mouth

THE MINOR GLANDS AND THEIR PRODUCT

Structure	What it is
Minor salivary glands	Small labial, buccal, palatal, and lingual glands scattered through the lining of the mouth
Saliva	The watery secretion that moistens, cleanses, and lubricates food and begins its breakdown

The Liver

The liver is the largest gland in the body. It sits below the diaphragm, mostly in the upper right abdomen, and it makes bile.

THE LOBES, LIGAMENTS, AND MICROSCOPIC STRUCTURES OF THE LIVER

Structure	What it is
Liver	The large gland below the diaphragm; it produces bile
Right and left lobes	The two principal lobes of the liver, separated by the falciform ligament
Quadrate and caudate lobes	Two smaller lobes on the posterior and inferior surface
Falciform ligament	The fold of peritoneum that suspends the liver from the anterior abdominal wall and diaphragm
Ligamentum teres	The round ligament; a fibrous cord in the free edge of the falciform ligament, a remnant of the fetal umbilical vein
Hepatocytes	The functional cells of the liver, arranged in plates called hepatic laminae
Hepatic sinusoids	The wide, leaky capillaries that run between the plates of hepatocytes
Bile canaliculi	The tiny channels between hepatocytes that collect the bile they make
Portal triad	The trio found at the corner of each liver lobule: a bile duct, a branch of the hepatic artery, and a branch of the hepatic portal vein

HOLD ONTO THIS

The portal triad is three things at the corner of every lobule, and the traffic runs both ways: the branch of the hepatic artery and the branch of the hepatic portal vein bring blood in, while the bile duct carries bile out.

The Gallbladder and the Path of Bile

The gallbladder is a small sac on the underside of the liver that stores bile. Bile travels a branching set of ducts to reach the small intestine.

THE GALLBLADDER AND THE BILE DUCTS

Structure	What it is
Gallbladder	The pear-shaped sac on the posterior surface of the liver that stores and concentrates bile
Fundus, body, and neck	The three regions of the gallbladder, from the broad inferior end to the tapered superior end
Cystic duct	The duct that carries bile to and from the gallbladder
Common bile duct	The duct formed where the common hepatic duct and the cystic duct join
Hepatopancreatic ampulla	The ampulla of Vater, the short widened duct where the bile duct and pancreatic duct meet before the duodenum

Bile drains away from the liver cells along a path of widening ducts to reach the duodenum.

1. **Bile canaliculi** tiny channels between hepatocytes collect the bile
2. **Hepatic ducts** the bile is gathered into the right and left hepatic ducts, which merge into the common hepatic duct
3. **Cystic duct and gallbladder** bile may detour into the gallbladder for storage through the cystic duct
4. **Common bile duct** the common hepatic and cystic ducts join to form the common bile duct
5. **Hepatopancreatic ampulla** the common bile duct joins the pancreatic duct at the ampulla
6. **Duodenum** bile enters the small intestine at the major duodenal papilla, past the sphincter of Oddi

The Pancreas

The pancreas lies behind the stomach. It is both an exocrine gland, secreting into the duodenum, and an endocrine gland.

THE REGIONS, DUCTS, AND CLUSTERS OF THE PANCREAS

Structure	What it is
Pancreas	The gland posterior to the stomach, retroperitoneal in position
Head, body, and tail	The three regions of the pancreas, from the wide end in the curve of the duodenum to the tapered end
Pancreatic duct	The main duct, also called the duct of Wirsung; it joins the common bile duct
Accessory duct	The smaller duct, also called the duct of Santorini, that empties separately into the duodenum
Major duodenal papilla	The raised opening in the duodenal lining where the hepatopancreatic ampulla delivers bile and pancreatic juice
Sphincter of Oddi	The muscular ring that controls flow through the hepatopancreatic ampulla into the duodenum
Acini	The exocrine clusters of the pancreas that produce pancreatic juice; about 99 percent of the gland
Pancreatic islets	The endocrine clusters of the pancreas that produce hormones; about 1 percent of the gland

HOLD ONTO THIS

Each pancreatic duct carries two names. The main one is the duct of Wirsung and the smaller one is the duct of Santorini, and only the main one joins the common bile duct.

Common Disorders of the Accessory Organs

Compare the common disorders of the accessory organs by the structure each one affects.

THE COMMON DISORDERS OF THE ACCESSORY ORGANS

Disorder	What it is
Gallstones	Hardened deposits that form in the gallbladder and can block the bile ducts
Cholecystitis	Inflammation of the gallbladder, often caused by a blocking gallstone
Hepatitis	Inflammation of the liver, commonly from a viral infection
Cirrhosis	The replacement of healthy liver tissue with scar tissue, with loss of liver function
Jaundice	A yellowing of the skin and eyes from a buildup of bilirubin, often from liver or bile duct disease
Pancreatitis	Inflammation of the pancreas
Mumps	A viral infection that inflames and enlarges the parotid glands

SHARED DUCTS, SHARED PROBLEMS

The bile duct and the pancreatic duct meet at the **hepatopancreatic ampulla** before they reach the duodenum. Because they share that final passage, a gallstone lodged at the ampulla can block bile and pancreatic juice at once, which is one way gallstones trigger pancreatitis.

The liver's blood supply is unusual. It receives nutrient-rich blood from the **hepatic portal vein** as well as oxygenated blood from the **hepatic artery**. That heavy blood flow filters the body's nutrients, but it also makes the liver a common site for cancer to spread to.

CHAPTER 5

Endocrine Anatomy

ENDOCRINE ANATOMY

COMPETENCIES 128 TO 132

Covers the endocrine glands, where each one sits in the body, and the gross anatomy of the pituitary, thyroid, parathyroid, adrenal, and pancreatic glands. Hormone action belongs to physiology and is not covered here.

BY THE END

1. Name the major endocrine glands and locate each one in the body.
2. Describe the gross anatomy of the pituitary gland and its connection to the hypothalamus.
3. Identify the parts of the thyroid, parathyroid, adrenal, and pancreatic glands.
4. Name organs that are not endocrine glands but still contain endocrine tissue.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Endocrine anatomy

drsrennie-stack.github.io/new-build-bio4-solano/endocrine-concept-videos.html

The Endocrine System, an Overview

The endocrine system is a scattered set of glands and tissues that signal the body by releasing hormones into the blood. Covers where those glands sit and how each one is built.

THE BASIC VOCABULARY

Structure	What it is
Endocrine system	The glands and tissues that release hormones into the blood to signal distant cells
Hormone	A chemical messenger released by an endocrine cell that travels in the blood to act on target cells
Endocrine gland	A ductless gland that secretes its product directly into the surrounding fluid and blood
Exocrine gland	By contrast, a gland that secretes through a duct onto a surface; not part of the endocrine system
Target cell	A cell that carries the receptor for a given hormone and therefore responds to it

The Major Endocrine Glands

The pure endocrine glands do nothing but make hormones. They are spread from the brain to the pelvis. Compare them by where each one sits.

WHERE EACH GLAND SITS

Gland	Location	Note
Hypothalamus	The floor and walls of the third ventricle, in the diencephalon of the brain	Links the nervous and endocrine systems and directs the pituitary
Pituitary gland	In the sella turcica of the sphenoid bone, hanging below the hypothalamus	The small gland the hypothalamus controls
Pineal gland	The posterior roof of the third ventricle, in the brain	A small, cone-shaped gland
Thyroid gland	The anterior neck, just below the larynx, wrapped around the trachea	The largest purely endocrine gland
Parathyroid glands	The posterior surface of the thyroid gland, usually four small glands	Tiny glands embedded behind the thyroid
Thymus	The mediastinum, behind the sternum	Large in childhood, shrinks after puberty
Adrenal glands	One capping the superior pole of each kidney	Also called the suprarenal glands
Pancreatic islets	Scattered through the pancreas, behind the stomach	The endocrine portion of the pancreas
Ovaries	The female pelvis, one on each side of the uterus	The female gonads
Testes	The scrotum	The male gonads

The Pituitary Gland

The pituitary gland, or hypophysis, is about the size of a pea. It sits in the sella turcica and hangs from the hypothalamus by a stalk. Its two lobes come from two different tissues.

THE PITUITARY AND ITS CONNECTIONS

Structure	What it is
Hypophysis	Another name for the pituitary gland
Sella turcica	The saddle-shaped pocket of the sphenoid bone that cradles and protects the pituitary
Infundibulum	The slender stalk that connects the pituitary to the hypothalamus above it
Hypophyseal portal system	The set of blood vessels that carries hypothalamic signals directly to the anterior lobe

The two lobes

Compare the two lobes by the tissue they are made of and how each connects to the hypothalamus.

THE LOBES OF THE PITUITARY GLAND

Lobe	Other name	Tissue	Link to the hypothalamus
Anterior lobe	Adenohypophysis	Glandular epithelial tissue	The hypophyseal portal system, a set of blood vessels
Posterior lobe	Neurohypophysis	Nervous tissue, an extension of the hypothalamus itself	A tract of nerve fibers running down through the infundibulum

HOLD ONTO THIS

The posterior lobe is nervous tissue, so it connects by nerve fibers. The anterior lobe is glandular epithelium, so it connects by blood vessels. The tissue tells you the link.

The Thyroid and Parathyroid Glands

The thyroid gland sits across the front of the trachea, and the four small parathyroid glands are embedded in its back surface.

The thyroid gland

THE THYROID, FROM THE GROSS GLAND TO THE FOLLICLE

Structure	What it is
Thyroid gland	The largest purely endocrine gland, two lateral lobes lying against the trachea below the larynx
Isthmus	The bridge of thyroid tissue, crossing the front of the trachea, that connects the two lobes
Thyroid follicles	The microscopic hollow spheres that make up the gland
Colloid	The protein-rich fluid stored in the center of each follicle
Follicular cells	The cells forming the wall of a follicle
Parafollicular cells	Also called C cells, scattered between the follicles

The parathyroid glands

THE PARATHYROID GLANDS AND THEIR CELLS

Structure	What it is
Parathyroid glands	Usually four small, round glands on the posterior surface of the thyroid lobes
Chief cells	The main hormone-producing cell of the parathyroid glands

The Adrenal Glands

An adrenal, or suprarenal, gland caps the top of each kidney. Each gland is really two glands in one: an outer cortex and an inner medulla, formed from different tissues.

THE TWO PARTS OF AN ADRENAL GLAND

Structure	What it is
Adrenal cortex	The outer region and the bulk of the gland, glandular tissue arranged in three layered zones
Adrenal medulla	The inner core, modified nervous tissue that belongs to the sympathetic nervous system

The cortex is built in three zones. Follow them in order, from the surface inward.

1. **Zona glomerulosa** the thin outer zone, just beneath the connective tissue capsule
2. **Zona fasciculata** the thick middle zone, the widest of the three, its cells in straight columns
3. **Zona reticularis** the inner zone, a branching network of cells against the medulla

The Pancreas and the Gonads

The pancreas and the gonads are mixed organs. Each does an endocrine job alongside another role, so only part of the organ is endocrine tissue.

THE MIXED ORGANS

Structure	What it is
Pancreas	A soft gland behind the stomach, both an exocrine and an endocrine organ
Pancreatic islets	Also called the islets of Langerhans, the small clusters of endocrine cells scattered through the pancreas
Acini	By contrast, the exocrine clusters of the pancreas that make digestive enzymes; not endocrine tissue
Ovaries	The paired female gonads in the pelvis, endocrine glands as well as reproductive organs
Testes	The paired male gonads in the scrotum, endocrine glands as well as reproductive organs

Organs with Endocrine Tissue

Beyond the pure endocrine glands, many organs whose main job is something else still carry pockets of endocrine cells. Compare them.

ORGANS THAT CONTAIN ENDOCRINE TISSUE

Organ	Endocrine tissue it contains
Hypothalamus	Neurosecretory cells; a neuroendocrine organ that controls the pituitary gland
Heart	Hormone-producing cells in the wall of the atria
Kidneys	Cells that release a hormone driving red blood cell production, plus an enzyme acting on blood pressure
Stomach and small intestine	Scattered endocrine cells in the lining of the digestive tract
Liver	Cells that contribute to a growth-related hormone and to blood pressure control
Skin	Cells that begin the production of vitamin D, which acts as a hormone
Adipose tissue	Fat cells that release a hormone signaling the body's energy stores
Placenta	A temporary organ of pregnancy that releases hormones supporting the pregnancy

WHERE A GLAND SITS SHAPES ITS DISEASE

The **pituitary gland** is boxed inside the bony **sella turcica**, so a pituitary tumor has nowhere to expand but upward, where it presses on the optic chiasm. A loss of peripheral vision can be the first sign that a gland the size of a pea has enlarged.

The **thyroid gland** lies just under the skin of the anterior neck, so an enlarged thyroid, a goiter, is visible and can be felt. The same superficial position lets a clinician palpate the gland and makes thyroid surgery accessible, though the tiny **parathyroid glands** on its back surface must be carefully spared.

Each **adrenal gland** rests on the superior pole of a kidney, which is why imaging ordered for the kidney also captures the adrenal, and why an adrenal mass is sometimes found by accident.

PART B

Lab Structure List.

Part B of the Module 4 packet. These are the lab structures and concepts to learn for the lab portion of the Module 4 exam. Cross reference them against every lab worksheet, and be prepared to identify structures on models, images, the cadaver, and slides.

1 **The Respiratory System**

2 **The Lymphatic System**

3 **The Alimentary Canal**

4 **The Accessory Digestive Organs**

5 **Vessels of the Abdomen and Pelvis**

6 **Muscles, Nerves and Vessels of the Lower Limb**

7 **The Endocrine System**

8 **The Anterior Neck**

CHAPTER 1

The Respiratory System

MODELS, CADAVER, AND SLIDES

COMPETENCIES 97 TO 104

Follow the air. Every structure in this list sits somewhere along one continuous path, and if you can trace the path you can place any structure on it. The thoracic block at the end is the same cadaver dissection the master list asks for, so work the two together.

☐ External nose and nasal cavity

STRUCTURES TO IDENTIFY

- ☐ Nares (nostrils)
- ☐ Nasal vestibule
- ☐ Nasal cavity
- ☐ Nasal septum
- ☐ Septal cartilage
- ☐ Perpendicular plate of the ethmoid
- ☐ Vomer
- ☐ Superior nasal concha
- ☐ Middle nasal concha
- ☐ Inferior nasal concha
- ☐ Hard palate
- ☐ Soft palate
- ☐ Choanae

☐ Paranasal sinuses

Name the bone each sinus is named for and say where it drains.

STRUCTURES TO IDENTIFY

- ☐ Frontal sinus
- ☐ Ethmoidal air cells
- ☐ Sphenoidal sinus
- ☐ Maxillary sinus

☐ Pharynx

Name the epithelium of each region and say whether it carries air, food, or both.

STRUCTURES TO IDENTIFY

- ☐ Nasopharynx
- ☐ Pharyngeal tonsil
- ☐ Opening of the auditory tube
- ☐ Oropharynx
- ☐ Palatine tonsil
- ☐ Lingual tonsil
- ☐ Laryngopharynx
- ☐ Uvula
- ☐ Soft palate

☐ Larynx

Identify which single cartilage is elastic rather than hyaline, and which one is the only complete ring.

STRUCTURES TO IDENTIFY

- ☐ Thyroid cartilage
- ☐ Laryngeal prominence
- ☐ Cricoid cartilage
- ☐ Epiglottis
- ☐ Arytenoid cartilages
- ☐ Corniculate cartilages
- ☐ Cuneiform cartilages
- ☐ Cricothyroid ligament
- ☐ Vocal folds (true vocal cords)
- ☐ Vocal ligaments
- ☐ Vestibular folds (false vocal cords)
- ☐ Glottis
- ☐ Hyoid bone

☐ Trachea and bronchial tree

Be prepared to say which main bronchus is wider and more vertical, and why that matters.

STRUCTURES TO IDENTIFY

- ☐ Trachea
- ☐ Tracheal cartilages (C rings)
- ☐ Trachealis muscle
- ☐ Carina
- ☐ Right main (primary) bronchus
- ☐ Left main (primary) bronchus
- ☐ Lobar (secondary) bronchi
- ☐ Segmental (tertiary) bronchi
- ☐ Bronchioles
- ☐ Terminal bronchioles

☐ Lungs and pleurae

STRUCTURES TO IDENTIFY

- ☐ Right lung, superior lobe
- ☐ Right lung, middle lobe
- ☐ Right lung, inferior lobe
- ☐ Horizontal fissure
- ☐ Oblique fissure
- ☐ Left lung, superior lobe
- ☐ Left lung, inferior lobe
- ☐ Cardiac notch
- ☐ Lingula
- ☐ Apex of the lung
- ☐ Base of the lung
- ☐ Hilum of the right lung
- ☐ Hilum of the left lung
- ☐ Parietal pleura
- ☐ Visceral pleura
- ☐ Right pleural cavity
- ☐ Left pleural cavity

☐ The thorax on the cadaver

Memory trick for the fiber directions: hands in pockets is external, praying hands is internal.

STRUCTURES TO IDENTIFY

- ☐ Thoracic cavity
- ☐ Mediastinum
- ☐ Esophagus
- ☐ Diaphragm
- ☐ External intercostals
- ☐ Internal intercostals

Innervation of the diaphragm. _____

Name the three structures that pass through the diaphragm and the cord level of each: T8 _____ T10 _____ T12 _____

☐ Vessels and nerves of the thorax

Give the side and say whether each vessel is an artery or a vein before you name it.

STRUCTURES TO IDENTIFY

- ☐ Brachiocephalic trunk
- ☐ Right subclavian artery
- ☐ Left subclavian artery
- ☐ Right common carotid artery
- ☐ Left common carotid artery
- ☐ Right phrenic nerve
- ☐ Left phrenic nerve
- ☐ Right brachiocephalic vein
- ☐ Left brachiocephalic vein
- ☐ Right subclavian vein
- ☐ Left subclavian vein
- ☐ Right internal jugular vein
- ☐ Left internal jugular vein
- ☐ Right pulmonary artery
- ☐ Left pulmonary artery
- ☐ Right pulmonary veins
- ☐ Left pulmonary veins

☐ The respiratory zone and the alveolus

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- ☐ Respiratory bronchiole
- ☐ Alveolar duct
- ☐ Alveolar sac
- ☐ Alveolus
- ☐ Pulmonary capillary

☐ Respiratory slides

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- ☐ Pseudostratified ciliated columnar epithelium
- ☐ Goblet cells
- ☐ Cilia
- ☐ Hyaline cartilage of the trachea
- ☐ Simple squamous epithelium of the alveolus
- ☐ Lung tissue, low power

☐ **Also identify**

STRUCTURES TO IDENTIFY

☐ Larynx ☐ Hilum ☐ Pleural cavities

Trace a breath of air from the nostril to an alveolus, naming every structure it passes.

CHAPTER 2

The Lymphatic System

MODELS, CADAVER, AND SLIDES

COMPETENCIES 105 TO 108

A one-way drainage network. The master list names the spleen and the three tonsils outright; the vessels, the node, and the organ list come from the lymphatic chapter of this module. Learn the route first and the parts fall onto it.

☐ **Lymphatic vessels, from the tissue spaces to the veins**

Say which duct drains the upper right quadrant and which drains the rest of the body.

STRUCTURES TO IDENTIFY

☐ Lymphatic capillaries ☐ Lymphatic trunks ☐ Subclavian veins
☐ Lacteals ☐ Right lymphatic duct ☐ Internal jugular veins
☐ Lymphatic collecting vessels ☐ Thoracic duct
☐ Lymphatic valves ☐ Cisterna chyli

☐ **The lymph node**

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

☐ Capsule ☐ Afferent lymphatic vessels ☐ Cervical nodes
☐ Cortex ☐ Efferent lymphatic vessels ☐ Axillary nodes
☐ Medulla ☐ Hilum ☐ Inguinal nodes

☐ **The lymphatic organs**

Sort each one as a primary or a secondary lymphatic organ.

STRUCTURES TO IDENTIFY

☐ Thymus ☐ Lymph nodes ☐ Spleen

☐ **The spleen**

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

☐ Spleen

☐ **The tonsils**

Name the region of the pharynx each tonsil guards.

STRUCTURES TO IDENTIFY

- ☐ Pharyngeal tonsil
- ☐ Palatine tonsil
- ☐ Lingual tonsil

Which two lymphatic organs are primary, and what happens in each?

Trace a drop of interstitial fluid from a tissue space back to the bloodstream.

CHAPTER 3

The Alimentary Canal

THE CADAVER, THE TORSO MODEL, AND SLIDES

COMPETENCIES 109 TO 119

One continuous tube from the mouth to the anus. Work the peritoneum first, because the folds are what you have to move aside to reach the organs, and then take the canal region by region.

☐ **Body cavities, serous membranes, and the omenta**

STRUCTURES TO IDENTIFY

- ☐ Abdominal cavity
- ☐ Pelvic cavity
- ☐ Peritoneal cavity
- ☐ Parietal peritoneum
- ☐ Visceral peritoneum
- ☐ Retroperitoneum
- ☐ Greater omentum
- ☐ Lesser omentum
- ☐ Falciform ligament
- ☐ Mesentery
- ☐ Mesocolon

Name the retroperitoneal structures. Memory device: SADPUCKER.

☐ **The wall of the canal, in section**

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- ☐ Mucosa
- ☐ Epithelium
- ☐ Lamina propria
- ☐ Muscularis mucosae
- ☐ Submucosa
- ☐ Muscularis, inner circular layer
- ☐ Muscularis, outer longitudinal layer
- ☐ Serosa
- ☐ Adventitia
- ☐ Lumen

☐ **Mouth, pharynx, and esophagus**

Say which end of the esophagus carries an adventitia rather than a serosa.

STRUCTURES TO IDENTIFY

- ☐ Oral cavity
- ☐ Hard palate
- ☐ Soft palate
- ☐ Uvula
- ☐ Palatine tonsil
- ☐ Lingual tonsil
- ☐ Nasopharynx
- ☐ Oropharynx
- ☐ Laryngopharynx
- ☐ Esophagus
- ☐ Esophageal hiatus
- ☐ Upper esophageal sphincter
- ☐ Lower (cardiac) esophageal sphincter

☐ The stomach

The stomach carries one more muscle layer than the rest of the tube. Name it and give its position.

STRUCTURES TO IDENTIFY

- ☐ Cardiac region
- ☐ Fundus
- ☐ Body
- ☐ Pyloric region
- ☐ Pyloric antrum
- ☐ Pyloric canal
- ☐ Pylorus
- ☐ Pyloric sphincter
- ☐ Cardiac sphincter
- ☐ Lesser curvature
- ☐ Greater curvature
- ☐ Gastric rugae

☐ The small intestine

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- ☐ Duodenum
- ☐ Major duodenal papilla
- ☐ Jejunum
- ☐ Ileum
- ☐ Ileocecal valve
- ☐ Ileocecal sphincter
- ☐ Mesentery
- ☐ Circular folds (plicae circulares)
- ☐ Villi
- ☐ Lacteal
- ☐ Microvilli (brush border)
- ☐ Sphincter of Oddi (hepatopancreatic ampulla)

☐ The large intestine

Name the three features that belong to the large intestine and to no other part of the canal.

STRUCTURES TO IDENTIFY

- ☐ Cecum
- ☐ Vermiform appendix
- ☐ Ascending colon
- ☐ Hepatic flexure (right colic flexure)
- ☐ Transverse colon
- ☐ Splenic flexure (left colic flexure)
- ☐ Descending colon
- ☐ Sigmoid colon
- ☐ Rectum
- ☐ Anal canal
- ☐ Internal anal sphincter
- ☐ External anal sphincter
- ☐ Teniae coli
- ☐ Haustra
- ☐ Omental appendices

☐ Also identify

STRUCTURES TO IDENTIFY

- ☐ Stomach
- ☐ Small intestine
- ☐ Large intestine

Trace a bite of food from the mouth to the anus, naming every region it passes through.

CHAPTER 4

The Accessory Digestive Organs

THE CADAVER, THE TORSO MODEL, AND SLIDES

COMPETENCIES 120 TO 125

The organs that sit alongside the tube and empty into it. Two questions on this sheet are hers outright: the fetal remnant behind the ligamentum teres, and the path of bile.

☐ The tongue and the salivary glands

Give the location of each of the three large salivary glands and say where its duct opens.

STRUCTURES TO IDENTIFY

- ☐ Tongue
- ☐ Lingual frenulum
- ☐ Papillae
- ☐ Parotid gland
- ☐ Submandibular gland
- ☐ Sublingual gland

☐ The liver

STRUCTURES TO IDENTIFY

- ☐ Right lobe
- ☐ Left lobe
- ☐ Quadrate lobe
- ☐ Caudate lobe
- ☐ Falciform ligament
- ☐ Ligamentum teres (round ligament)
- ☐ Lesser omentum

Ligamentum teres. It is the remnant of which fetal structure? _____

☐ Liver tissue, on the slide and the model

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- ☐ Hepatocytes
- ☐ Hepatic sinusoids
- ☐ Bile canaliculi
- ☐ Portal triad
- ☐ Bile duct
- ☐ Branch of the hepatic artery
- ☐ Branch of the hepatic portal vein
- ☐ Central vein

☐ The gallbladder and the biliary tree

STRUCTURES TO IDENTIFY

- ☐ Gallbladder
- ☐ Fundus of the gallbladder
- ☐ Body of the gallbladder
- ☐ Neck of the gallbladder
- ☐ Right hepatic duct
- ☐ Left hepatic duct
- ☐ Common hepatic duct
- ☐ Cystic duct
- ☐ Common bile duct
- ☐ Hepatopancreatic ampulla
- ☐ Major duodenal papilla
- ☐ Sphincter of Oddi

☐ The pancreas

Say which part of the pancreas is exocrine and which is endocrine, and which duct joins the common bile duct.

STRUCTURES TO IDENTIFY

- ☐ Head of the pancreas
- ☐ Body of the pancreas
- ☐ Tail of the pancreas
- ☐ Main pancreatic duct
- ☐ Accessory pancreatic duct
- ☐ Hepatopancreatic ampulla
- ☐ Major duodenal papilla
- ☐ Acini

☐ Also identify

STRUCTURES TO IDENTIFY

- ☐ Pancreas
- ☐ Liver

Trace a drop of bile from the hepatocyte that made it to the duodenum.

CHAPTER 5

Vessels of the Abdomen and Pelvis

CADAVER AND TORSO MODEL

COMPETENCY 126

The master list groups these as arteries then veins, with the hepatic portal system kept separate. Learn the three unpaired branches of the abdominal aorta in order from the top, because their order is what the tag is testing.

☐ Arteries of the abdomen

Name the three unpaired visceral branches of the abdominal aorta in order from superior to inferior, and say what each one supplies.

STRUCTURES TO IDENTIFY

- ☐ Abdominal aorta
- ☐ Celiac trunk
- ☐ Left gastric artery
- ☐ Splenic artery
- ☐ Common hepatic artery
- ☐ Superior mesenteric artery
- ☐ Renal arteries
- ☐ Gonadal (testicular or ovarian) arteries
- ☐ Inferior mesenteric artery
- ☐ Right common iliac artery
- ☐ Left common iliac artery

☐ Arteries of the pelvis

STRUCTURES TO IDENTIFY

- ☐ Internal iliac artery
- ☐ External iliac artery
- ☐ Medial umbilical ligament

☐ Veins of the abdomen and pelvis

STRUCTURES TO IDENTIFY

- ☐ Inferior vena cava
- ☐ Renal veins
- ☐ Gonadal (testicular or ovarian) veins
- ☐ Right common iliac vein
- ☐ Left common iliac vein
- ☐ Internal iliac vein
- ☐ External iliac vein

☐ The hepatic portal system

Say what makes this a portal system, and name the two capillary beds the blood passes through.

STRUCTURES TO IDENTIFY

- ☐ Hepatic portal vein
- ☐ Splenic vein
- ☐ Superior mesenteric vein
- ☐ Inferior mesenteric vein
- ☐ Hepatic veins

Trace a molecule of glucose absorbed by the small intestine from the intestinal capillary to the right atrium.

CHAPTER 6

Muscles, Nerves and Vessels of the Lower Limb

MODELS AND CADAVER

COMPETENCY 127

Muscles first, then the nerves and vessels that run with them. For every muscle, be able to name it, find it on the models and the cadaver, and give its origin, insertion, and action. Use right and left, and artery and vein, when you name a vessel.

☐ Anterior, medial and lateral thigh

Give the origin, insertion, and action of each muscle.

STRUCTURES TO IDENTIFY

- ☐ Iliopsoas (iliacus and psoas major)
- ☐ Rectus femoris
- ☐ Vastus lateralis
- ☐ Vastus intermedius
- ☐ Vastus medialis
- ☐ Sartorius
- ☐ Gracilis
- ☐ Adductor magnus
- ☐ Adductor longus
- ☐ Adductor brevis
- ☐ Pectineus
- ☐ Tensor fasciae latae
- ☐ Iliotibial (IT) band
- ☐ Iliacus
- ☐ Psoas major
- ☐ Quadriceps femoris

☐ Gluteal region and posterior thigh

Give the origin, insertion, and action of each muscle.

STRUCTURES TO IDENTIFY

- ☐ Gluteus maximus
- ☐ Gluteus medius
- ☐ Biceps femoris
- ☐ Semitendinosus
- ☐ Semimembranosus
- ☐ Hamstrings

☐ The leg

Give the origin, insertion, and action of each muscle.

STRUCTURES TO IDENTIFY

- ☐ Tibialis anterior
- ☐ Extensor hallucis longus
- ☐ Extensor digitorum longus
- ☐ Fibularis brevis
- ☐ Fibularis longus
- ☐ Tibialis posterior
- ☐ Flexor digitorum longus
- ☐ Flexor hallucis longus
- ☐ Gastrocnemius
- ☐ Soleus
- ☐ Plantaris

☐ Nerves and vessels of the lower limb

STRUCTURES TO IDENTIFY

- ☐ Femoral nerve
- ☐ Sciatic nerve
- ☐ Tibial nerve
- ☐ Common fibular nerve
- ☐ Femoral artery
- ☐ Femoral vein
- ☐ Great saphenous vein
- ☐ Posterior tibial artery

CHAPTER 7

The Endocrine System

TORSO MODEL, CADAVER, AND SLIDES

COMPETENCIES 128 TO 132

The endocrine glands are scattered, so this sheet is a tour of locations first. The master list names the thyroid gland in the neck and the endocrine pancreas and the suprarenal glands in the abdomen; the rest come from the endocrine chapter of this module.

☐ Locating the glands

Give the location of each gland before you name it.

STRUCTURES TO IDENTIFY

- ☐ Hypothalamus
- ☐ Pituitary gland
- ☐ Pineal gland
- ☐ Thyroid gland
- ☐ Parathyroid glands
- ☐ Thymus
- ☐ Adrenal (suprarenal) glands
- ☐ Pancreas
- ☐ Ovaries
- ☐ Testes

☐ **The thyroid and parathyroid glands**

STRUCTURES TO IDENTIFY

- ☐ Thyroid gland, right lobe
- ☐ Thyroid gland, left lobe
- ☐ Isthmus
- ☐ Parathyroid glands
- ☐ Trachea
- ☐ Larynx

☐ **Thyroid tissue, on the slide**

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- ☐ Thyroid follicles
- ☐ Colloid
- ☐ Follicular cells
- ☐ Parafollicular (C) cells

☐ **The adrenal gland**

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- ☐ Adrenal (suprarenal) gland
- ☐ Adrenal cortex
- ☐ Zona glomerulosa
- ☐ Zona fasciculata
- ☐ Zona reticularis
- ☐ Adrenal medulla
- ☐ Kidney

☐ **The endocrine pancreas**

Say which of these two clusters is endocrine and which is exocrine.

STRUCTURES TO IDENTIFY

- ☐ Pancreas
- ☐ Pancreatic islets
- ☐ Acini

Which gland sits on the superior pole of each kidney, and what are its two parts made of?

_____

CHAPTER 8

The Anterior Neck

MODELS AND CADAVER

These are on the Unit 4 worksheets of the master list.

☐ **Anterior neck**

Platysma and sternocleidomastoid are on Exam 3 this time.

STRUCTURES TO IDENTIFY

- ☐ Sternocleidomastoid
- ☐ Common carotid artery
- ☐ Vertebral artery
- ☐ Internal jugular vein